

Triangle Properties — Practice Questions

Digital SAT Style | Mixed Difficulty | Total Questions: 25

Q1.

DIFFICULTY: EASY

In triangle ABC, angle A = 50° and angle B = 65° . What is the measure of angle C?

- A) 55°
 - B) 65°
 - C) 70°
 - D) 75°
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Q2.

DIFFICULTY: EASY

In triangle PQR, angle P = $(2x + 10)^\circ$, angle Q = $(3x - 5)^\circ$, and angle R = $(x + 25)^\circ$. What is the value of x?

- A) 20
 - B) 25
 - C) 30
 - D) 35
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Q3.

DIFFICULTY: EASY

An exterior angle of a triangle measures 130° . One of the non-adjacent interior angles measures 55° . What is the other non-adjacent interior angle?

- A) 50°
 - B) 65°
 - C) 75°
 - D) 80°
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Q4.

DIFFICULTY: EASY

In triangle ABC, angle A = angle B and angle C = 80° . What is the measure of angle A?

- A) 40°
 - B) 50°
 - C) 60°
 - D) 80°
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Q5.

DIFFICULTY: EASY

A right triangle has legs of length 6 cm and 8 cm. What is the length of the hypotenuse?

- A) 7 cm
 - B) 10 cm
 - C) 12 cm
 - D) 14 cm
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Q6.

DIFFICULTY: EASY

Triangle DEF is similar to triangle GHI. $DE = 4$, $EF = 6$, and $GH = 10$. What is HI?

- A) 12
 - B) 15
 - C) 18
 - D) 20
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Q7.

DIFFICULTY: EASY

In triangle XYZ, angle $X = 90^\circ$, angle $Y = 35^\circ$. What is angle Z?

- A) 45°
 - B) 55°
 - C) 65°
 - D) 90°
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Q8.

DIFFICULTY: EASY

The sides of a triangle are 5, 12, and 13. Is this a right triangle?

- A) Yes, because $5^2 + 12^2 = 13^2$
 - B) No, because $5 + 12 \neq 13^2$
 - C) Yes, because $5 + 12 = 17 > 13$
 - D) No, because 13 is odd
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Q9.

DIFFICULTY: MEDIUM

In triangle ABC, angle $A = 42^\circ$ and $AB = BC$ (isosceles). What is angle C?

- A) 42°
 - B) 69°
 - C) 96°
 - D) 138°
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Q10.

DIFFICULTY: MEDIUM

In triangle DEF, $DE = 9$, $EF = 12$, and $DF = 15$. What type of triangle is this?

- A) Acute
 - B) Right
 - C) Obtuse
 - D) Equilateral
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Q11.

DIFFICULTY: MEDIUM

Two similar triangles have areas in the ratio 9:25. What is the ratio of their corresponding sides?

- A) 3:5
 - B) 9:25
 - C) 9:5
 - D) 81:625
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Q12.

DIFFICULTY: MEDIUM

The midsegment of a triangle connects the midpoints of two sides. If one side of the triangle measures 18 cm, what is the length of the midsegment parallel to that side?

- A) 6 cm
 - B) 9 cm
 - C) 12 cm
 - D) 18 cm
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Q13.

DIFFICULTY: MEDIUM

In triangle ABC, angle $B = 90^\circ$. If $AB = 5$ and $BC = 12$, what is AC ?

- A) 7
 - B) 10
 - C) 13
 - D) 17
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Q14.

DIFFICULTY: MEDIUM

Which set of three lengths can form a triangle?

- A) 2, 3, 6
 - B) 1, 4, 6
 - C) 4, 5, 8
 - D) 3, 3, 7
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Q15.

DIFFICULTY: MEDIUM

In similar triangles ABC and DEF, $AB/DE = 3/4$. If the perimeter of triangle ABC is 36 cm, what is the perimeter of triangle DEF?

- A) 27 cm
 - B) 48 cm
 - C) 64 cm
 - D) 108 cm
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Q16.

DIFFICULTY: MEDIUM

In triangle PQR, angle P = $(3x + 10)^\circ$ and the exterior angle at R = $(7x - 30)^\circ$. Angle Q = $(2x)^\circ$. What is x?

- A) 15
 - B) 20
 - C) 25
 - D) 30
-

Q17.

DIFFICULTY: HARD

In triangle ABC, $AB = 8$, $AC = 6$, and $BC = 10$. What is the area of triangle ABC?

- A) 20
 - B) 24
 - C) 30
 - D) 40
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Q18.

DIFFICULTY: HARD

An equilateral triangle has a side length of 10. What is its height?

- A) 5
 - B) $5\sqrt{2}$
 - C) $5\sqrt{3}$
 - D) $10\sqrt{3}$
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Q19.

DIFFICULTY: HARD

The sides of a triangle are 13, 14, and 15. Using Heron's formula, what is the area?

- A) 72
 - B) 84
 - C) 90
 - D) 96
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Q20.

DIFFICULTY: HARD

Triangle ABC has vertices A(0, 0), B(6, 0), and C(3, 4). What is the area?

- A) 8
 - B) 10
 - C) 12
 - D) 16
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Q21.

DIFFICULTY: HARD

In triangle ABC, D is the midpoint of BC. If AB = 10, AC = 14, and BC = 16, what is the length of median AD?

- A) 8
 - B) 9
 - C) $\sqrt{58}$
 - D) $\sqrt{65}$
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Q22.

DIFFICULTY: HARD

Two triangles are similar. The sides of the smaller triangle are 5, 7, and 9. The longest side of the larger triangle is 27. What is the perimeter of the larger triangle?

- A) 42
 - B) 56
 - C) 63
 - D) 81
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Q23.

DIFFICULTY: HARD

In right triangle ABC with angle C = 90° , altitude CD is drawn to hypotenuse AB. If AD = 4 and DB = 9, what is CD?

- A) 5
 - B) 6
 - C) 6.5
 - D) 7
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Q24.

DIFFICULTY: HARD

In triangle ABC, the angle bisector from A meets BC at D. If AB = 12, AC = 8, and BC = 10, what is BD?

- A) 4

- B) 5
 - C) 6
 - D) 7
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Q25.

DIFFICULTY: HARD

A ladder 13 feet long leans against a wall. The foot of the ladder is 5 feet from the wall. How high up the wall does the ladder reach?

- A) 8 ft
 - B) 10 ft
 - C) 12 ft
 - D) 14 ft
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